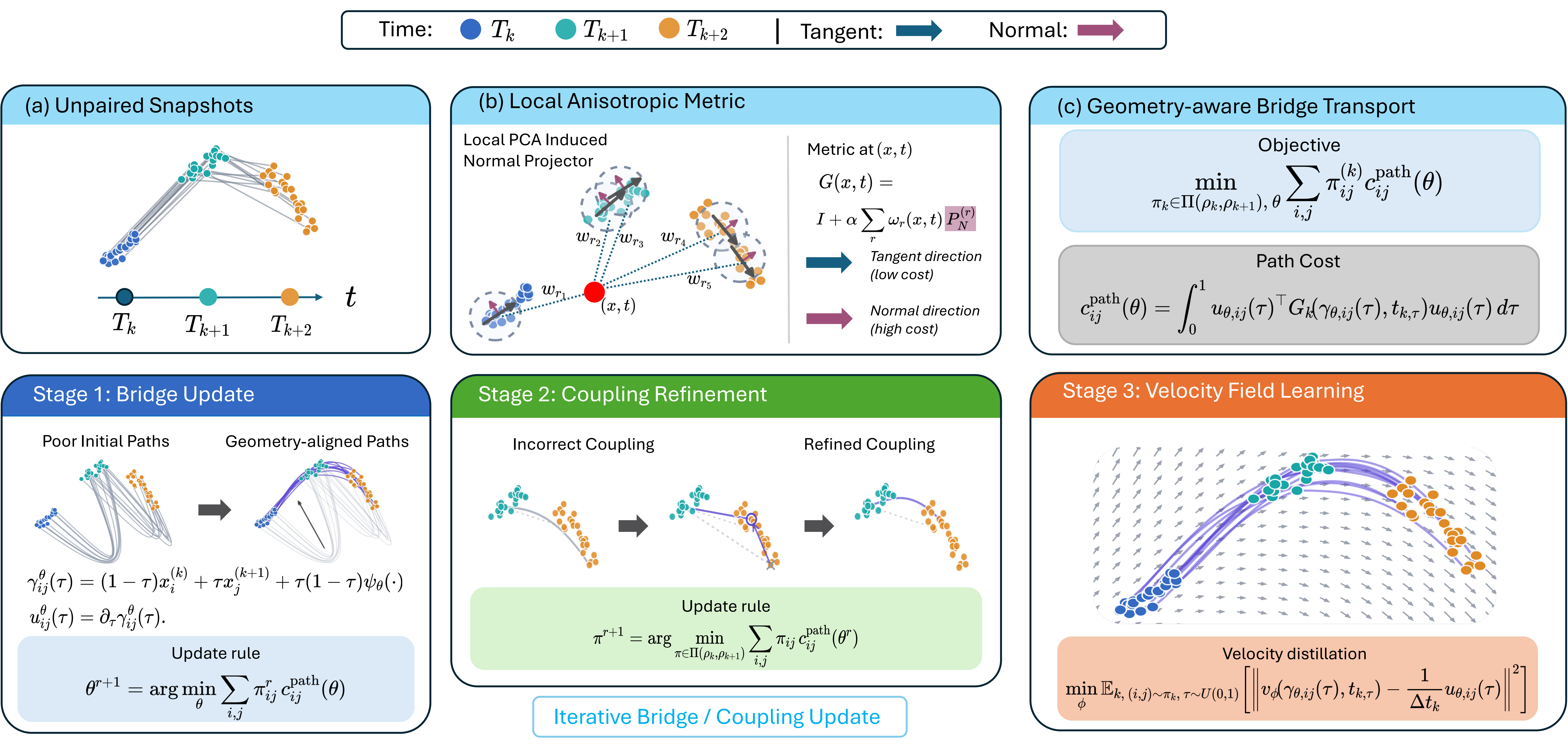

Time: $T_k$ $T_{k+1}$ $T_{k+2}$ | Tangent: Normal:
(a) Unpaired Snapshots
$T_k$ $T_{k+1}$ $T_{k+2}$ $t$
(b) Local Anisotropic Metric
Local PCA Induced Normal Projector
$w_{r_1}$ $w_{r_2}$ $w_{r_3}$ $w_{r_4}$ $w_{r_5}$ $(x,t)$
Metric at $(x,t)$
$G(x,t) = I + \alpha \sum_r \omega_r(x,t)\, P_N^{(r)}$
Tangent direction (low cost)
Normal direction (high cost)
(c) Geometry-aware Bridge Transport
Objective
$\min_{\pi_k \in \Pi(\rho_k,\rho_{k+1}),\, \theta} \sum_{i,j} \pi_{ij}^{(k)} c_{ij}^{\text{path}}(\theta)$
Path Cost
$c_{ij}^{\text{path}}(\theta) = \int_0^1 u_{\theta,ij}(\tau)^\top G_k(\gamma_{\theta,ij}(\tau), t_{k,\tau}) u_{\theta,ij}(\tau)\, d\tau$
Stage 1: Bridge Update
Poor Initial Paths
Geometry-aligned Paths
$\gamma_{ij}^{\theta}(\tau) = (1-\tau)x_i^{(k)} + \tau x_j^{(k+1)} + \tau(1-\tau)\psi_\theta(\cdot)$
$u_{ij}^{\theta}(\tau) = \partial_\tau \gamma_{ij}^{\theta}(\tau).$
Update rule
$\theta^{r+1} = \arg\min_\theta \sum_{i,j} \pi_{ij}^r c_{ij}^{\text{path}}(\theta)$
Stage 2: Coupling Refinement
Incorrect Coupling
Refined Coupling
Update rule
$\pi^{r+1} = \arg\min_{\pi \in \Pi(\rho_k,\rho_{k+1})} \sum_{i,j} \pi_{ij} c_{ij}^{\text{path}}(\theta^r)$
Iterative Bridge / Coupling Update
Stage 3: Velocity Field Learning
Velocity distillation
$\min_\phi \mathbb{E}_{k,\,(i,j)\sim\pi_k,\,\tau\sim U(0,1)} \left[ \left\| v_\phi(\gamma_{\theta,ij}(\tau), t_{k,\tau}) - \frac{1}{\Delta t_k} u_{\theta,ij}(\tau) \right\|^2 \right]$